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Contents

Page

Foreword	iv
Introduction	v
1 Scope	1
2 Normative references	1
3 Terms and definitions	1
4 General requirements	1
4.1 General	1
4.2 Validation and verification of methods	2
4.3 Equipment and consumables	3
4.4 Calibration and performance checks	3
4.5 Metrological traceability	3
4.6 Reference and control samples, collections and databases	4
4.7 Significant figures and measurement uncertainty	4
4.8 Performance monitoring	4
5 Personnel	5
6 Facilities	5
7 Environmental conditions	5
8 Acceptance and rejection of requests and items	5
8.1 Acceptance and rejection of the customer's request	5
8.2 Recording of accepted items	6
9 Analytical strategy	6
9.1 General requirements	6
9.2 Assessment prior to analysis	6
9.3 Selection of methods	7
9.4 Item selection, sampling and preparation	7
10 Purpose of the analysis	7
10.1 Classification/Identification	7
10.2 Quantification	8
10.3 Question of source	8
10.4 Reconstruction	8
11 Reliability of observations	9
Annex A (informative) The forensic process	10
Annex B (informative) Examples	12
Bibliography	14

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043- 2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043- 3, *Forensic Sciences – Part 3: Analysis*

ISO 21043- 4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043- 5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements designed to safeguard the process for the analysis of items. The requirements are designed to ensure the use of suitable methods, proper controls, qualified personnel, and appropriate analytical strategies throughout the forensic analysis of items.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A](#) and [B](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 3: Analysis

1 Scope

This document specifies requirements and provides recommendations for the analysis of items of potential forensic value. It includes requirements and recommendations for the selection and application of suitable method(s) for the analysis to meet the needs of the customer and fulfil the request.

It is applicable to activities conducted by a forensic service provider that occur at the scene and within a facility.

The requirements facilitate the comprehensive, accurate, and reliable analysis of items.

This document is applicable to all disciplines of forensic science; however, it is not applicable to the recovery of digital data which is covered by ISO/IEC 27037^[1].

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 21043-2, *Forensic sciences — Part 2: Recognition, recording, collecting, transport and storage of items*

Numerous requirements listed in this standard are also described in ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories*.

However, references to ISO/IEC 17025:2017 are for information purposes only [2].

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

4.1 General

The procedures, including methods, for the analysis of items shall be documented, suitable for their intended use and, when applicable, include information regarding:

- a) measures that mitigate - the risk of loss, degradation, contamination and/or alteration of items;

ISO/DIS 21043-3:2024(en)

- b) sampling/selection of items;
- c) item preparation;
- d) selection of methods suitable to address the customer's request;
- e) use of reference and control samples;
- f) metrological traceability;
- g) assessment of measurement uncertainty;
- h) activities to demonstrate reliability of the observations;
- i) maintenance, operation, and calibration of equipment (hardware and software), including relevant performance parameters that can substantially impact the observations and their interpretation;
- j) environmental conditions that can substantially impact the observations and their interpretation;
- k) method limitations;
- l) use of any external service providers.

The examiner shall record the relevant observations, data, information and methods used during analysis. Information includes reference to uniquely identified equipment and consumables that can substantially impact the reliability of the observations.

Deviations from a procedure shall be recorded and the record retained.

Refer to ISO 21043-2 for further requirements for item handling and control, including collecting, packaging, labelling, transport, storage, and chain of custody.

4.2 Validation and verification of methods

Methods, including software, calculations and measurements, shall be validated prior to implementation. Methods previously validated elsewhere shall be verified at the facility where they are to be used, prior to implementation. Off-the-shelf analysis software used within its intended purpose and scope may be considered sufficiently validated, but shall be verified prior to implementation.

Known source items, comparable to those encountered in casework, shall be used for method validation and verification.

When changes are made to a validated method, the influence of such changes shall be determined and where they are found to affect the original validation, a new method validation shall be performed prior to implementation^[2].

Method validation should include the following performance characteristics, when applicable:

- a) precision;
- b) accuracy;
- c) limit of detection;
- d) limit of quantitation;
- e) specificity/selectivity;
- f) calibration;
- g) robustness;
- h) carry-over/contamination.

ISO/DIS 21043-3:2024(en)

Records of method validation and verification shall be retained, e.g., in the form of a validation plan and a validation report. These shall include:

- i) procedure used;
- j) specification of the requirements;
- k) determination of the performance characteristics of the method;
- l) observations obtained;
- m) statement on the validity of the method, detailing its suitability for the intended use, relevant limitations of the method and caveats about its use;
- n) person who performed the validation or verification.

NOTE See ISO/IEC 17025:2017, 7.2 *Selection, verification and validation of methods* for guidance.

4.3 Equipment and consumables

All equipment that can substantially impact the results shall be uniquely identified and records on the version of the software and firmware shall be retained, if applicable. Equipment shall be maintained and operated as per manufacturer's instruction or the forensic service provider's documented procedure.

Consumables that can substantially impact the observations shall be checked against pre-established acceptance criteria for proper/intended function before use. Where consumables form part of a batch supply, then batch testing where appropriate may be used. The checks shall be recorded and the record retained.

NOTE The frequency of the check can depend on the type of consumable and frequency of use.

Refer to ISO 21043-2 for further requirements for equipment and consumables.

4.4 Calibration and performance checks

Equipment and software that can substantially impact the observations shall be checked against pre-established acceptance criteria for proper/intended function prior to use and shall be periodically checked to support its continued use.

Records of calibration and/or performance checks shall be retained. These records shall include:

- a) the unique identity of the equipment;
- b) the version of the software and firmware used, if applicable;
- c) the identity of any reference or control samples used;
- d) the date of the calibration and/or performance checks and the identity of the personnel performing them;
- e) the results of the calibration and/or performance checks and the corresponding evaluation;
- f) actions taken when calibration and/or performance checks are outside the expected outcome.

4.5 Metrological traceability

When necessary for the reliability of observations, metrological traceability of measurements shall be established through appropriate calibration of measuring equipment and/or use of certified reference materials.

NOTE See ISO/IEC 17025:2017, 6.5 *Metrological Traceability* for guidance.

ISO/DIS 21043-3:2024(en)

4.6 Reference and control samples, collections and databases

The forensic service provider shall, when applicable, utilize reference samples, control samples, collections or databases in the analysis of items.

Reference and control samples shall be:

- a) uniquely identified; and
- b) traceable to certified reference materials or obtained from known sources.

The origin of samples or the relevant known characteristics of samples in collections and databases shall be documented.

Reference samples, control samples and collections shall be stored in a manner that mitigates the risk of loss, degradation, contamination or alteration.

Databases shall be stored in a manner that mitigates the risk of loss and unauthorized alteration.

Reference or control samples shall not be used beyond their expiry date, unless it has been checked that they remain fit for purpose beyond this date. The results of the check shall be retained.

4.7 Significant figures and measurement uncertainty

The forensic service provider shall have a documented procedure to estimate the measurement uncertainty of reported measurement observations. Estimation of measurement uncertainty shall be included in the validation of quantitative methods.

When estimating measurement uncertainty, all components of sampling and analysis that can substantially impact measurement uncertainty shall be taken into account. The major sources of uncertainty relating to a method shall be documented along with the overall uncertainty and an explanation of how the overall uncertainty was derived.

The forensic service provider shall have a documented procedure addressing the use of significant figures and the rounding of numerical values.

4.8 Performance monitoring

The forensic service provider shall have a documented procedure that addresses the monitoring of its performance which may include:

- a) proficiency testing;
- b) interlaboratory comparison;
- c) intralaboratory comparison.

Performance monitoring can be declared or blind.

The forensic service provider shall participate regularly in a relevant proficiency testing program or interlaboratory comparison, if available. Performance monitoring programs should include sets of examples typical of casework complexity. If an appropriate proficiency test or interlaboratory comparison does not exist, the forensic service provider should implement an intralaboratory comparison.

The forensic service provider should choose proficiency testing providers that meet the requirements of ISO/IEC 17043. The frequency of participation shall be based on risk and shall be sufficient to demonstrate continued performance and competence.

When participating in performance monitoring, the forensic service provider shall use the same procedures as it does for casework. The criteria specifying what constitutes sufficient performance shall be defined prior to the monitoring.

ISO/DIS 21043-3:2024(en)

The results of performance monitoring shall be compared to the expected results or pre-established criteria and, when there are significant deviations, action(s) shall be taken to address the deviations.

Performance monitoring records shall include:

- d) identifiers of personnel involved and how they were involved;
- e) the identity of the equipment used;
- f) an identifier for the performance monitoring test undertaken;
- g) the analyses undertaken and the observations obtained;
- h) an evaluation against expected observations or pre-established criteria;
- i) any corrective action(s) undertaken;
- j) the evaluation of any impact on previous and current analysis where necessary.

Performance monitoring should also include analysis of trends if multiple performance tests are undertaken.

5 Personnel

Roles and responsibilities of personnel shall be defined and documented. Personnel involved in the forensic process, including external providers engaged by the forensic service provider, shall be competent to perform their role.

Refer to ISO 21043-2 for further requirements for personnel.

6 Facilities

Facilities shall be suitable for their intended use. Facilities shall be accessible to authorized personnel only.

The forensic service provider shall have a documented procedure to mitigate the risk of contamination. Procedures shall include maintaining a clean environment to the extent necessary for the analysis.

Incompatible activities shall be separated (e.g., suitable separation of pre- and post-PCR activities within a DNA facility).

Refer to ISO 21043-2 for further requirements for facilities.

7 Environmental conditions

If environmental conditions (e.g., temperature, humidity, magnetic field) can substantially impact the reliability of the observations, they shall be monitored, recorded, and, if possible, controlled.

The forensic service provider shall have documented procedure(s) that address how to manage relevant environmental conditions including what actions to take if any of these go out of specification.

Equipment shall operate reliably in the environmental conditions in which it is used.

8 Acceptance and rejection of requests and items

8.1 Acceptance and rejection of the customer's request

The forensic service provider shall have a documented procedure to review and record the customer's request which should include:

- a) criteria for acceptance or rejection of the customer's request;

ISO/DIS 21043-3:2024(en)

b) criteria for acceptance or rejection of items.

The criteria for accepting or rejecting a submitted request or submitted items for analysis shall be made available to the customers.

The request shall be reviewed for clarity to ensure the items are relevant to the request.

The customer's request for service shall be recorded.

The forensic service provider shall consult with the customer when the written request for service is ambiguous.

In instances where the request is rejected, the reason for the rejection shall be made available to the customer and the records should be retained.

8.2 Recording of accepted items

The acceptance of items shall be recorded. The rejection of items should be recorded. The reason for rejection of items should be recorded and be available to the customer.

The forensic service provider shall record any conditions regarding the packaging or integrity of items that can negatively affect the suitability for analysis. Where there is doubt about suitability of an item or when the item does not conform to the description provided, the forensic service provider should consult with the customer to clarify the issues and record the outcome.

Once accepted, if not already appropriately labelled and/or packaged, items shall be labelled, packaged and stored in accordance with ISO 21043-2. Where the integrity of an item has been compromised prior to analysis, measures shall be taken to preserve the item from further degradation, contamination, or alteration. The condition of the item and the measures taken shall be recorded.

9 Analytical strategy

9.1 General requirements

The analytical strategy shall take into consideration:

- a) prioritization of the items;
- b) task-relevant information;
- c) the nature, quality and relevance of the items;
- d) the capabilities and limitations of the available methods; and
- e) the appropriateness and benefits of the additional testing of items. In order to allow re-analysis when necessary, the analytical strategy should give consideration to retaining a portion of the item.

If the item will be fully consumed during analysis the customer should be notified prior to analysis. If the risk of fully consuming the item is always present for a type of analysis, and the customer has been informed about this risk, the request may be considered as implicit approval.

The forensic service provider shall have a documented procedure to address the common sequences of analysis.

9.2 Assessment prior to analysis

The forensic service provider shall assess whether or not analysis of the items would address the customer's request in the context of the information/case circumstances provided. A description of the items shall be recorded prior to their analysis.

ISO/DIS 21043-3:2024(en)

If the items are not sufficient to address the customer's request, the forensic service provider should consult with the customer regarding the limitations of the analysis given the items received. The forensic service provider should record the outcome of the consultation.

When analyses will be performed despite item suitability being questioned, this should be recorded.

9.3 Selection of methods

The selection of analytical methods shall be based on their suitability for the intended use and should consider:

- a) the purpose of the analysis (see [section 10](#));
- b) the nature of the item to be analyzed;
- c) the quantity of the item available for analysis;
- d) the method(s) available;
- e) the non-destructive or destructive nature of the method(s);
- f) the availability of suitable equipment and operators;
- g) the sensitivity and specificity;
- h) the accuracy required;
- i) the measurement uncertainty.

Where the quantity of an item is limited or the analysis is likely to consume or alter the item, the sequence of analysis shall be planned and implemented to ensure that non-destructive methods are given priority and conducted before any destructive methods are attempted.

Any deviation from a validated or verified analysis method shall be technically justified and recorded. The expected impact of the deviation on the validity of the method shall be recorded.

9.4 Item selection, sampling and preparation

The forensic service provider shall have documented procedures for item selection, sampling and preparation for each type of item that is routinely analyzed.

NOTE 1 Sampling of items is carried out for either or both of the following reasons:

- a) to obtain material in a form suitable for analysis;
- b) to obtain a representative sample of the item received.

Sampling methods shall take into account how representativeness is obtained when only part of an item is analyzed. When homogenization is necessary for the representativeness of the sample, but is not possible, several samples of the item should be analyzed separately.

NOTE 2 Recommendation for drug sampling strategies can be found in reference UN *Guidelines on Representative Drug Sampling*, United Nations Office on Drugs and Crime, ENFSI Drugs Working Group, 2011^[3].

10 Purpose of the analysis

10.1 Classification/Identification

The forensic service provider shall have a documented method for the comparison of items to relevant reference samples, collections and databases.

ISO/DIS 21043-3:2024(en)

Analysis for classification shall be based on the specificity of any features used and on the suitability of the database or the taxonomic key used.

Annex [B.1](#) includes examples of classification.

Methods that are used to identify classes of compounds and that have a high rate of false positive observations as a consequence of low specificity, shall be treated as presumptive only.

Analysis for identification shall use at least one method of high selectivity when available.

In the identification process, positive observations from methods of low specificity shall be confirmed using one or more methods of higher specificity, if methods of higher specificity are available.

Criteria required for identification shall be established for methods used.

NOTE Identification is a special case of classification.

Annex [B.2](#) includes examples of identification.

10.2 Quantification

Quantification methods shall have acceptance criteria that include, at minimum:

- a) control sample requirements;
- b) requirements for acceptable calibration.

The forensic service provider shall have a documented procedure to address when replicate quantitative testing is needed. When replicate testing is performed, at least two samples from the original item shall be analyzed, and the variation between the calculated quantities shall be within defined limits.

For any analytical method applied to determine the concentration or purity of a compound(s) of interest, the limits and the working range of the calibration curve shall be defined. Defined acceptance criteria shall require the quantitative value to be within the working range of the calibration curve and shall prescribe actions to take when a quantitative value is outside of this range.

If multiple observations from controls are determined to fall outside of the acceptance criteria, the method shall be checked, and the inconsistencies recorded and the record retained.

Annex [B.3](#) includes examples of quantification.

10.3 Question of source

Where analysis for question of source involves the comparison of an item from an unknown source with an item from a known source, the item from the known source shall be suitable for the required comparison, and shall be selected to be as representative as possible of the known source.

Annex [B.4](#) includes examples of questions of source.

10.4 Reconstruction

Analyses for reconstruction are carried out to address whether an observed effect could have been the consequence of a particular activity. Analyses for reconstruction should be based on empirical data and may involve experimental simulations under controlled conditions. If relevant to the case circumstances, such analyses shall consider whether alternative experimental conditions produce the same observed effect.

Annex [B.5](#) includes examples of reconstruction.

11 Reliability of observations

The forensic service provider shall have a documented procedure that includes checking of:

- a) analytical observations;
- b) data transfers;
- c) comparisons;
- d) measurements;
- e) manual calculations.

The procedure shall specify at what stage in analysis checks occur, who conducts checks, and what methods require checks to be performed by a second examiner competent in the same area of expertise. The forensic service provider shall have a documented procedure to address disputes arising from the check of analysis. A record of checks shall be retained.

Annex A **(informative)**

The forensic process

The figure below is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

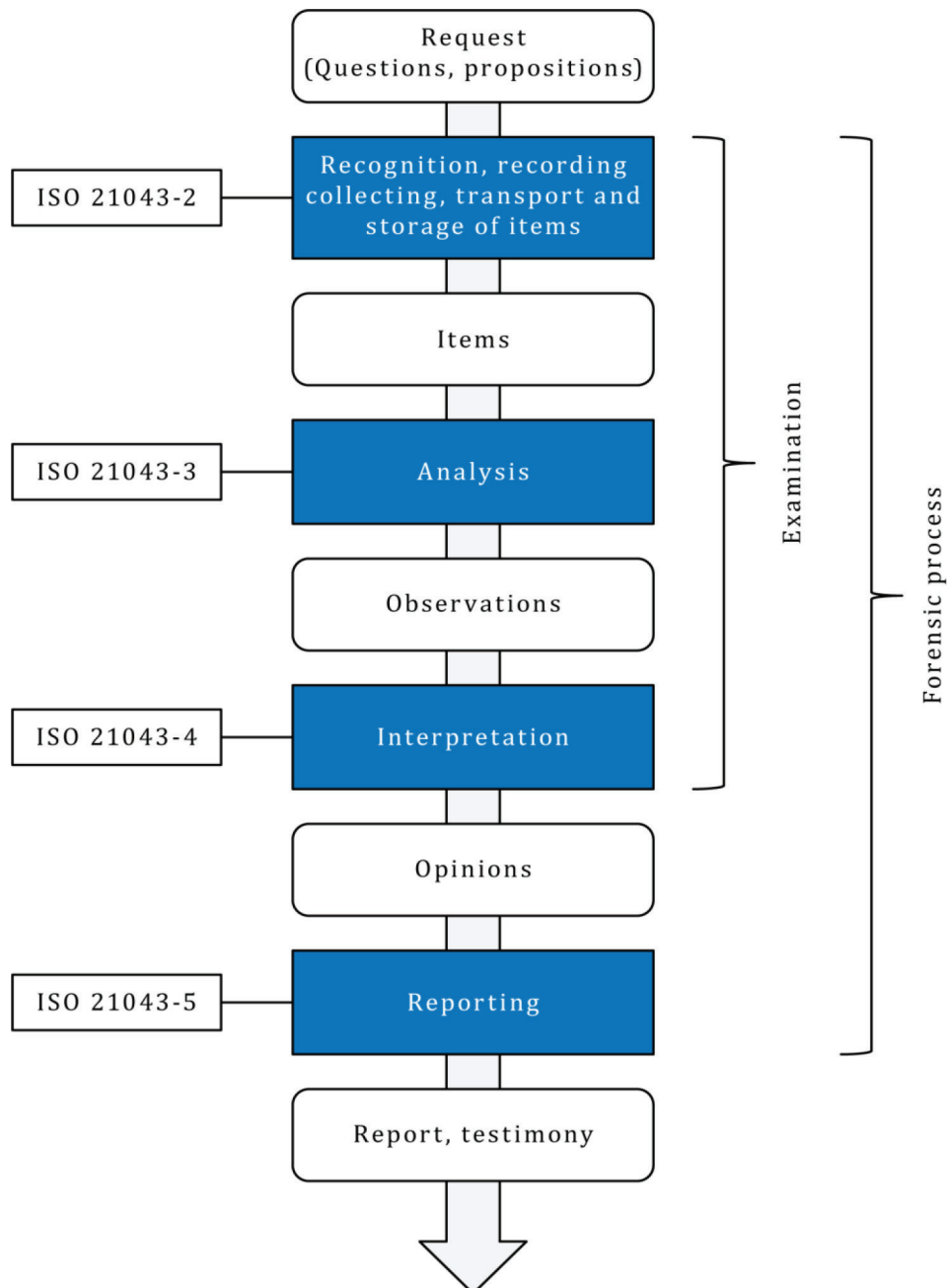


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

Examples

B.1 Examples of classification

- EXAMPLE 1 Determining the manufacturer of a motor vehicle based on the comparison of a paint item from the crime scene with a motor vehicle paint database.
- EXAMPLE 2 Determining the manufacturer of a cartridge case, based on a comparison with a physical ammunition collection or a searchable database of cartridges from known manufacturers.
- EXAMPLE 3 Comparing a footwear mark (unknown) to a piece of footwear (known), to infer potential manufacturer.
- EXAMPLE 4 Analysing the properties of an impression and classify them as a finger, palm, or shoe impression.
- EXAMPLE 5 Comparing the properties of a bloodstain pattern of unknown type with the properties of bloodstain pattern of known type to determine the type of pattern.
- EXAMPLE 6 Comparing an unknown piece of malware found on a computer with a collection of pieces of malware representing known classes of malware.

B.2 Examples of identification

- EXAMPLE 1 Identifying human blood on an item through serological testing.
- EXAMPLE 2 In drug analysis, comparing the spectrum of an unknown item to that in a known drug reference database to identify the type of drug.
- EXAMPLE 3 Comparison between the RNA of an unknown type of cell present on an item with the RNAs from known types of cells.
- EXAMPLE 4 Identifying cannabis based on microscopic morphological properties.

B.3 Examples of quantification

- EXAMPLE 1 Determining the concentration of ethanol (alcohol) in blood.
- EXAMPLE 2 Measuring the quantity or the purity of cocaine in a white powder mixture.
- EXAMPLE 3 Determining the weight measurement of a suspected controlled substance.
- EXAMPLE 4 In digital casework, determining the number of illegal images in the total number of downloaded images.

B.4 Examples of questions of source

- EXAMPLE 1 Comparing a fingerprint found on an item (unknown source) to a reference fingerprint of a known person (known source).
- EXAMPLE 2 Comparing microscopic and chemical characteristics of carpet fibers recovered from the clothing of a dumped body (unknown source) to the interior carpet fiber in a suspect vehicle (known source).
- EXAMPLE 3 Comparing a footwear mark (unknown source) to a specific shoe (known source).
- EXAMPLE 4 Comparing the DNA found on an item (unknown source) with the DNA reference sample from a specific individual (known source).

ISO/DIS 21043-3:2024(en)

EXAMPLE 5 Comparing the handwriting on a document (unknown source) with reference samples of handwriting from a specific individual (known source).

EXAMPLE 6 Comparing the noise pattern from a digital picture (unknown source) with the noise pattern from digital pictures originating from the smartphone of a specific individual (known source).

B.5 Examples of reconstruction

EXAMPLE 1 Examining bloodstain patterns to determine possible activities that led to them.

EXAMPLE 2 Measuring features (e.g., position of cartridge cases, trajectories and impact locations of bullets) at a crime scene, to address the chain of events and activities leading to these observations.

EXAMPLE 3 Determining the presence of specific features in a file system, file sharing applications, and in digital images to address the chain of events leading to the presence of illegal images on a computer.

Bibliography

- [1] ISO/IEC 27037, *Information technology — Security techniques — Guidelines for identification, collection, acquisition and preservation of digital evidence*
- [2] ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories*
- [3] UN *Guidelines on Representative Drug Sampling*, United Nations Office on Drugs and Crime, ENFSI Drugs Working Group, 2011